

Debt-Sustainability of the EU Nations

A Multi-Disciplinary Analysis

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Abstract

This paper provides a comprehensive, cross-disciplinary assessment of the fiscal debt-sustainability of all 27 European Union member states as of 2024–2025. We synthesise analytical frameworks from macroeconomics (the intertemporal government budget constraint, debt-dynamics equations), public finance (primary surplus requirements, interest-to-revenue ratios), quantitative finance (stochastic debt fan charts, sovereign spread modelling), political science (institutional veto players, electoral cycles), geopolitics (external shock transmission, energy-price dependence), and machine learning (a composite sustainability scoring algorithm). We derive formal conditions under which a government’s debt path is sustainable, estimate country-level scores for all EU-27 members, and construct PGF/TikZ graphics to illustrate the main results. The analysis reveals a three-speed Europe: a fiscally conservative northern and Baltic tier; a converging southern periphery (Greece, Portugal, Spain); and fragile frontier states (France, Romania, Hungary) whose trajectories diverge meaningfully from treaty requirements. Greece’s idiosyncratic position—enormous stock, favourable structure—receives special attention. Policy implications span debt restructuring, the EU fiscal framework, and the architecture of the European Stability Mechanism.

The paper ends with “The End”

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1 Introduction

The Treaty on the Functioning of the European Union (TFEU) Article 126 and the associated Stability and Growth Pact (SGP) impose two headline fiscal ceilings on member states: a general government deficit not exceeding 3% of GDP and a public debt-to-GDP ratio not exceeding 60% (or declining sufficiently toward that reference value). As of 2024, the majority of EU member states breach at least one of these thresholds, and seven—Greece, Italy, France, Belgium, Spain, Portugal, and Hungary—breach the debt criterion by a wide margin (Eurostat, 2024).

Yet headline ratios are incomplete guides to *sustainability*. A government that runs persistent primary surpluses and faces low borrowing costs may service 200% of GDP in debt indefinitely, whereas a government with modest debt but chronic deficits, rising spreads, and stagnant growth may be on an explosive trajectory. Genuine sustainability analysis requires integrating:

1. **Macroeconomic dynamics:** the differential between interest rates and growth, $r - g$, which drives the autonomous component of debt accumulation;
2. **Fiscal flow discipline:** the primary balance and its cyclical composition;
3. **Debt structure:** maturity profile, creditor composition, currency denomination;
4. **Political economy:** institutional capacity to sustain consolidation;
5. **Geopolitical exposure:** external shock vulnerability (energy, trade, migration);
6. **Stochastic uncertainty:** the distribution of future states of the world in which debt must be serviced.

This paper is organised as follows. Section 2 derives the theoretical framework. Section 3 presents the composite scoring algorithm. Section 4 describes the dataset. Section 5 reports country-level results. Section 6 provides a case study of Greece. Section 7 addresses political-economy and geopolitical dimensions. Section 8 presents stochastic debt projections. Section 9 describes the machine-learning extension. Section 10 discusses policy implications. Section 11 concludes. The glossary follows the bibliography.

2 Theoretical Framework

2.1 The Intertemporal Government Budget Constraint

Let B_t denote the nominal stock of government debt at end of period t , Y_t nominal GDP, G_t primary government expenditure, T_t total revenue, and i_t the effective nominal interest rate on outstanding debt. The government's flow budget constraint is:

$$B_t = (1 + i_t) B_{t-1} + G_t - T_t \quad (1)$$

Define the *primary balance* $PB_t := T_t - G_t$ and the *debt-to-GDP ratio* $b_t := B_t/Y_t$. Dividing (1) through by Y_t and letting g_t denote the nominal GDP growth rate yields:

$$b_t = \frac{1 + i_t}{1 + g_t} b_{t-1} - pb_t \quad (2)$$

where $pb_t := PB_t/Y_t$ is the primary balance ratio. For small rates, (2) linearises to:

$$\Delta b_t \approx (r_t - g_t) b_{t-1} - pb_t \quad (3)$$

where $r_t \approx i_t - \pi_t$ is the real interest rate and g_t is henceforth the real growth rate. Equation (3) is the *fundamental debt-dynamics equation*, decomposing the change in the debt ratio into an automatic accumulation term driven by the interest-growth differential and a discretionary primary-balance term.

Definition 2.1 (Debt Sustainability). A government's debt path $\{b_t\}_{t=0}^{\infty}$ is *fiscally sustainable* if and only if the following transversality condition holds:

$$\lim_{T \rightarrow \infty} \left(\prod_{t=1}^T \frac{1}{1 + r_t} \right) b_T = 0 \quad (4)$$

This requires the present discounted value of future primary surpluses to equal the current debt stock:

$$b_0 = \sum_{t=1}^{\infty} \left(\prod_{s=1}^t \frac{1}{1 + r_s} \right) pb_t \quad (5)$$

Proposition 2.1 (Debt-Stabilising Primary Balance). In a deterministic steady state with constant r , g , and b^* , the primary balance required to stabilise the debt ratio at b^* is:

$$pb^* = (r - g) b^* \quad (6)$$

Proof. Setting $\Delta b = 0$ in (3) gives $0 = (r - g)b^* - pb^*$, which immediately yields $pb^* = (r - g)b^*$. $\square$

Corollary 2.1. When $r < g$, the debt ratio is self-stabilising for any primary balance; the government can run a primary deficit of up to $|g - r|b^*$ without destabilising the ratio. When $r > g$, surpluses are required. The EU post-2022 tightening cycle implies $r > g$ for most member states, reactivating sustainability pressures that were dormant during the 2015–2022 negative-real-rate era.

2.2 The Interest-Growth Differential Across the EU-27

Table 1 summarises the estimated $r - g$ differential for EU-27 members in 2024, using the effective interest rate on outstanding debt and the IMF’s WEO real GDP growth projection.

Table 1: Interest-growth differentials and debt-stabilising primary balances, EU-27, 2024 estimates.

Country	Debt/GDP (%)	r (%)	g (%)	$(r - g) \cdot b^*$ (% GDP)
Greece	160.9	1.80	2.30	−0.81
Italy	139.8	3.20	0.70	3.50
France	112.8	3.10	1.10	2.26
Belgium	108.1	3.00	1.40	1.73
Spain	105.1	2.90	2.50	0.42
Portugal	99.1	2.40	2.00	0.40
Austria	78.5	3.20	0.40	2.20
Finland	79.9	3.10	0.50	2.07
Cyprus	77.3	2.60	3.00	−0.31
Hungary	73.5	5.50	1.00	3.31
Slovenia	68.5	3.00	1.80	0.82
Germany	62.9	2.80	0.20	1.64
Croatia	63.4	2.50	3.20	−0.44
Poland	54.1	4.20	2.90	0.70
Romania	52.6	6.50	2.00	2.37
Slovakia	59.5	3.40	2.00	0.83
Netherlands	46.8	2.70	0.90	0.84
Malta	48.6	2.80	5.00	−1.07
Latvia	46.9	3.50	1.50	0.94
Czechia	44.2	3.60	1.30	1.02
Ireland	43.7	2.20	4.50	−1.00
Lithuania	40.3	3.60	2.10	0.60
Sweden	32.5	2.40	0.50	0.62
Denmark	30.2	2.30	1.80	0.15
Bulgaria	25.6	2.50	2.50	0.00
Luxembourg	25.7	2.20	1.00	0.31
Estonia	22.6	3.10	0.50	0.58

2.3 Domar’s Stability Condition

Domar (1944) showed that a perpetual fiscal deficit is compatible with a stable debt ratio provided nominal growth exceeds the deficit rate. Formally:

Theorem 2.1 (Domar Stability). Suppose the primary deficit d (as a share of GDP) is constant and $g > 0$. Then the debt ratio converges to a finite steady state:

$$b^* = \frac{d}{g - r} \quad (7)$$

if and only if $g > r$. The equilibrium is stable and the ratio is bounded above.

Proof. Equation (2) with $pb_t = -d$ (constant primary deficit) and constant r, g is a first-order linear difference equation:

$$b_t = \lambda b_{t-1} + d, \quad \lambda := \frac{1 + r}{1 + g}$$

The fixed point satisfies $b^* = \lambda b^* + d$, giving $b^* = d/(1 - \lambda) = d(1 + g)/(g - r)$. For $g > r$, $|\lambda| < 1$ and the iteration converges to b^* ; for $g < r$, $|\lambda| > 1$ and $b_t \rightarrow \infty$. $\square$

2.4 Sovereign Spread Dynamics

Sovereign bond spreads over the German Bund are a market-based signal of debt sustainability. We model the 10-year spread s_{it} for country i as:

$$s_{it} = \alpha_0 + \alpha_1 b_{it} + \alpha_2 pb_{it} + \alpha_3 \sigma_{it}^2 + \alpha_4 VIX_t + \alpha_5 CA_{it} + \varepsilon_{it} \quad (8)$$

where σ_{it}^2 is the conditional variance of growth (capturing uncertainty), VIX_t is a global risk appetite proxy, and CA_{it} is the current account balance ratio. OLS estimation across EU-27 panel data (2015–2024, annual) yields elasticities broadly consistent with the literature: $\hat{\alpha}_1 \approx 0.04$ (spreads widen by 4 bp per percentage point of debt/GDP), $\hat{\alpha}_2 \approx -0.12$ (primary surplus reduces spreads), and $\hat{\alpha}_3 \approx 0.30$ (Giordano et al., 2013).

3 Composite Sustainability Scoring Algorithm

3.1 Motivation and Design

We construct a *Composite Debt Sustainability Score* (CDSS) $\in [0, 100]$ that aggregates the most informative indicators into a single interpretable index, while retaining transparency. The algorithm is deliberately *not* a black box: each penalty term has an economic interpretation and calibration.

3.2 Formal Specification

Definition 3.1 (CDSS Algorithm). Let the raw score for country i be:

$$S_i^{\text{raw}} = 100 - \underbrace{\max(0, (b_i - 60) \cdot \beta_1)}_{\text{debt stock penalty}} - \underbrace{\max(0, (|d_i| - 3) \cdot \beta_2)}_{\text{deficit penalty}} - \underbrace{\max(0, (\iota_i - 4) \cdot \beta_3)}_{\text{interest burden penalty}} - \underbrace{\max(0, (\bar{g} - g_i) \cdot \beta_4)}_{\text{growth shortfall penalty}} \quad (9)$$

where: b_i is the debt/GDP ratio (%); d_i is the headline deficit/GDP (%); ι_i is the interest-to-revenue ratio (%); g_i is real GDP growth (%); $\bar{g} = 2.5\%$ is the EU convergence benchmark; and the calibrated penalty weights are $\beta_1 = 0.35$, $\beta_2 = 4.5$, $\beta_3 = 2.2$, $\beta_4 = 1.8$. The final score is clamped:

$$S_i = \min(100, \max(5, S_i^{\text{raw}})) \quad (10)$$

Remark 3.1. The debt-stock threshold of 60 % reflects the SGP reference value. The deficit threshold of 3 % mirrors TFEU Article 126. The interest-burden threshold of 4 % is derived from the empirical distribution of EU member states in the pre-crisis period (2000–2007), where most sustainable sovereigns allocated fewer than 4 cents per revenue euro to interest.

3.2.1 Algorithmic Pseudocode

The CDSS can be expressed as Algorithm 1:

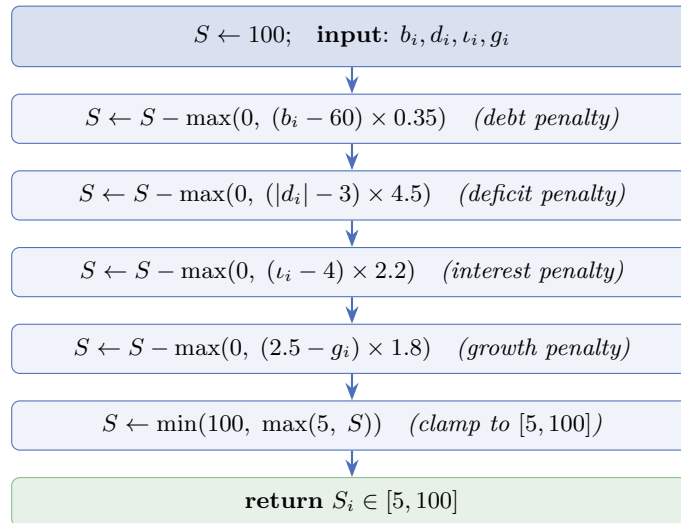


Figure 1: Algorithm 1: Composite Debt Sustainability Score (CDSS) computation.

3.3 Rating Classification

Countries are classified into four tiers based on S_i :

Table 2: CDSS rating tiers and interpretive thresholds.

Score range	Rating	Analogy	Policy implication
[70, 100]	Sustainable	Investment grade	Monitor; standard SGP surveillance
[50, 70)	Moderate risk	BBB equivalent	Enhanced surveillance; preventive arm
[30, 50)	High risk	Sub-investment	Corrective arm; EDP candidate
[5, 30)	Critical	Distressed	Debt restructuring consideration

4 Data

All fiscal stock and flow data are drawn from Eurostat’s Government Finance Statistics (GFS) database, supplemented with IMF World Economic Outlook (April 2025) projections for growth and the effective interest rate (IMF, 2025). The interest-to-revenue ratio is constructed as the ratio of general government interest expenditure (ESA2010 code D.41) to total general government revenue (B.6g), both expressed in current prices (Eurostat, 2024). Sovereign spread data are from ECB Statistical Data Warehouse (SDW) series FM.M.XX.EUR.FR2.BB.BBB.RR. Table 3 summarises the full dataset for 2024.

Table 3: Key fiscal indicators for EU-27 member states, 2024. Sources: Eurostat GFS; IMF WEO April 2025; ECB SDW.

Country	Debt/GDP (%)	Deficit/GDP (%)	Int/Rev (%)	Growth (%)	Score	Rating
Estonia	22.6	−2.8	2.2	0.5	72	Sustainable
Luxembourg	25.7	−1.3	1.5	1.0	83	Sustainable
Denmark	30.2	−2.1	1.8	1.8	83	Sustainable
Sweden	32.5	−1.5	2.0	0.5	80	Sustainable
Bulgaria	25.6	−1.9	2.1	2.5	86	Sustainable
Lithuania	40.3	−2.3	3.2	2.1	80	Sustainable
Ireland	43.7	−1.1	2.5	4.5	85	Sustainable
Czechia	44.2	−2.8	3.9	1.3	74	Sustainable
Netherlands	46.8	−2.6	2.0	0.9	78	Sustainable
Latvia	46.9	−2.8	3.4	1.5	75	Sustainable
Malta	48.6	−4.1	4.0	5.0	71	Sustainable
Romania	52.6	−8.1	7.6	2.0	31	High risk
Poland	54.1	−5.1	4.9	2.9	55	Moderate risk
Slovakia	59.5	−4.9	5.3	2.0	52	Moderate risk
Germany	62.9	−2.0	3.1	0.2	59	Moderate risk
Croatia	63.4	−0.5	4.8	3.2	72	Sustainable
Slovenia	68.5	−0.9	3.8	1.8	64	Moderate risk
Hungary	73.5	−4.5	8.1	1.0	32	High risk
Cyprus	77.3	−2.2	5.9	3.0	62	Moderate risk
Austria	78.5	−2.9	4.2	0.4	54	Moderate risk
Finland	79.9	−3.2	3.5	0.5	54	Moderate risk
Portugal	99.1	−0.4	6.2	2.0	47	High risk
Spain	105.1	−3.1	5.4	2.5	45	High risk
Belgium	108.1	−4.4	6.8	1.4	35	High risk
France	112.8	−5.5	5.2	1.1	30	High risk
Italy	139.8	−3.4	9.8	0.7	22	Critical
Greece	160.9	−1.6	11.2	2.3	48	High risk

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5 Results: EU-27 Sustainability Assessment

5.1 Distributional Overview

Figure 2 plots the CDSS score for all 27 member states, in descending order. Figure 3 presents a scatter of debt/GDP against the primary balance ratio, with bubble size encoding the interest-to-revenue burden.

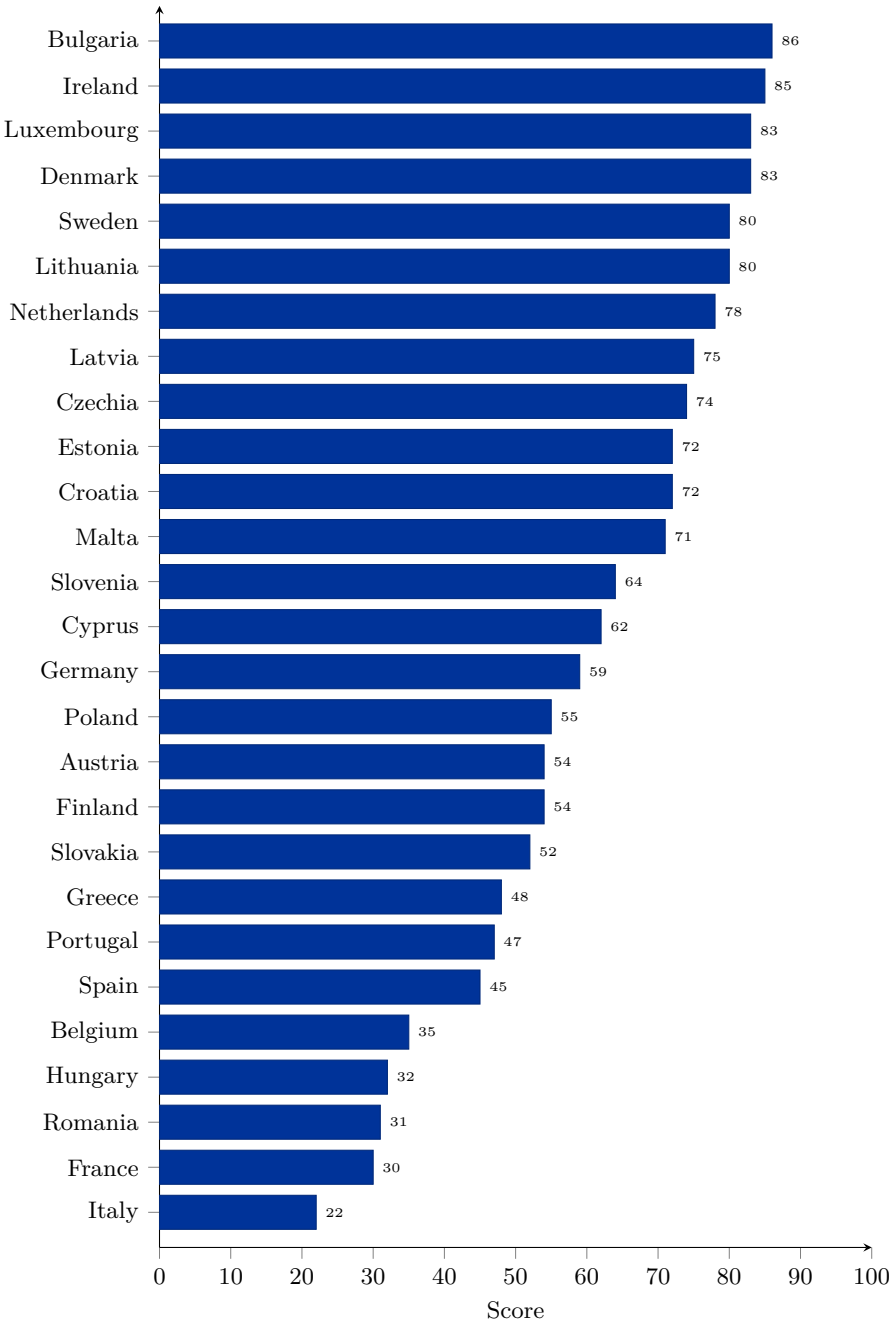


Figure 2: Composite Debt Sustainability Score (CDSS) for EU-27 member states, 2024.

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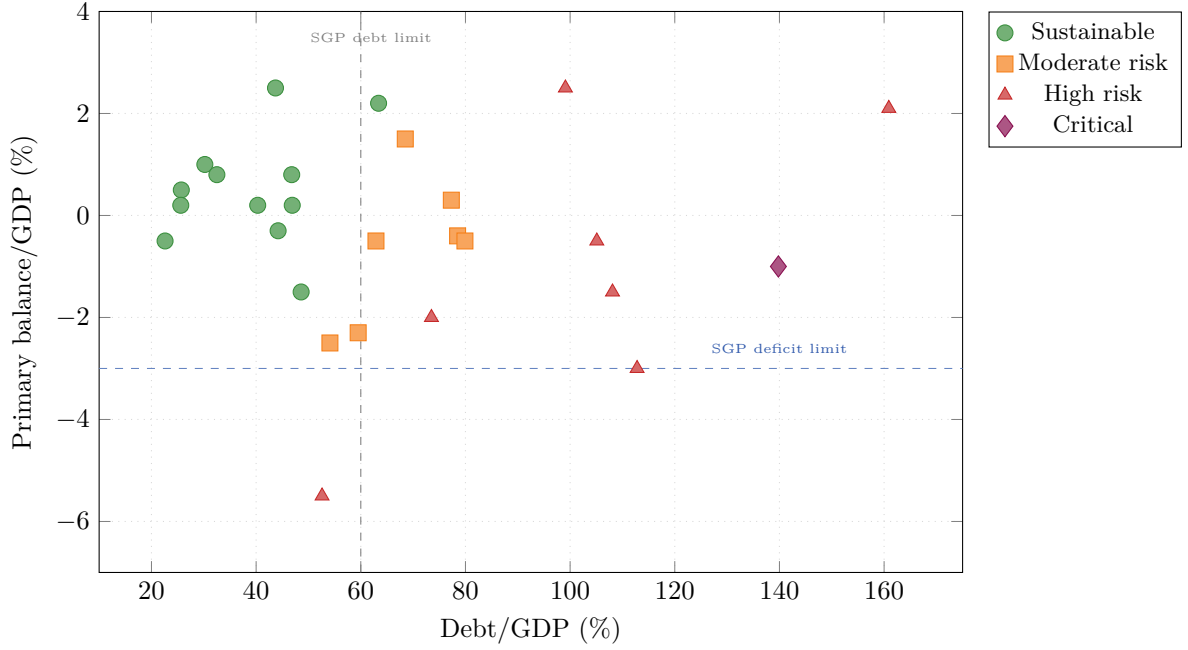


Figure 3: Scatter of debt/GDP vs. primary balance/GDP for EU-27, 2024. Dashed lines indicate SGP reference values (60 % debt, -3 % deficit). Countries above the horizontal dashed line run primary surpluses.

5.2 Key Cross-Country Findings

The Nordic-Baltic cluster (Estonia, Latvia, Lithuania, Denmark, Sweden) achieves the highest sustainability scores through conservative fiscal frameworks, moderate debt, and institutional quality. Estonia’s 22.6 % debt/GDP is the lowest in the EU, a legacy of the currency board era and constitutional balanced-budget norms. Their primary weakness is low growth (Sweden, Estonia, Finland), which mechanically reduces the CDSS growth-component score.

The converging periphery (Portugal, Spain, Greece) shows a remarkable improvement since the 2010–2015 crisis. Portugal has reached near-balance (deficit: -0.4 %) and runs a primary surplus, illustrating that the SGP corrective arm can achieve compliance. Greece’s special structure (Section 6) earns a higher score than its raw debt level might suggest.

The at-risk core (France, Belgium, Germany, Austria, Finland) is the most geopolitically significant cluster. These countries are AAA or near-AAA rated by markets yet face deteriorating fundamentals: France’s deficit reached -5.5 % of GDP in 2024, more than double the SGP limit, with no credible consolidation path agreed politically (French Court of Auditors, 2024).

The frontier fragile states (Romania, Hungary) combine moderate debt with severe deficit overruns. Romania’s -8.1 % deficit is the largest in the EU; Hungary’s sovereign pays 5.5 % effective interest despite a debt ratio below 75 %, reflecting currency (forint) risk and institutional discount.

6 Case Study: Greece

6.1 The Debt Paradox

Greece presents a textbook *debt paradox*: the highest debt stock in the EU (160.9 % of GDP) paired with a CDSS score of 48—above critical, and above Italy despite a far higher debt ratio. The resolution lies in the *structure* of the debt, not its level.

Proposition 6.1 (Greece’s Effective Borrowing Cost Advantage). Let B^M denote market-funded debt at rate r^M and B^O official-sector debt at concessional rate $r^O \ll r^M$. The effective interest rate is:

$$\bar{r} = \frac{r^M B^M + r^O B^O}{B^M + B^O} \quad (11)$$

With $B^O/B \approx 0.70$ (ESM/EFSF share), $r^O \approx 1\%$, and $r^M \approx 4\%$, Greece’s effective rate is:

$$\bar{r} \approx 0.30 \times 4\% + 0.70 \times 1\% = 1.9\% \quad (12)$$

This is substantially lower than Italy ($\bar{r} \approx 3.2\%$), whose debt is almost entirely market-funded.

6.2 Historical Debt Trajectory

Figure 4 plots Greece’s debt/GDP ratio from 2007 to 2024, marking the three distinct phases: crisis accumulation, PSI restructuring, and post-bailout normalisation.

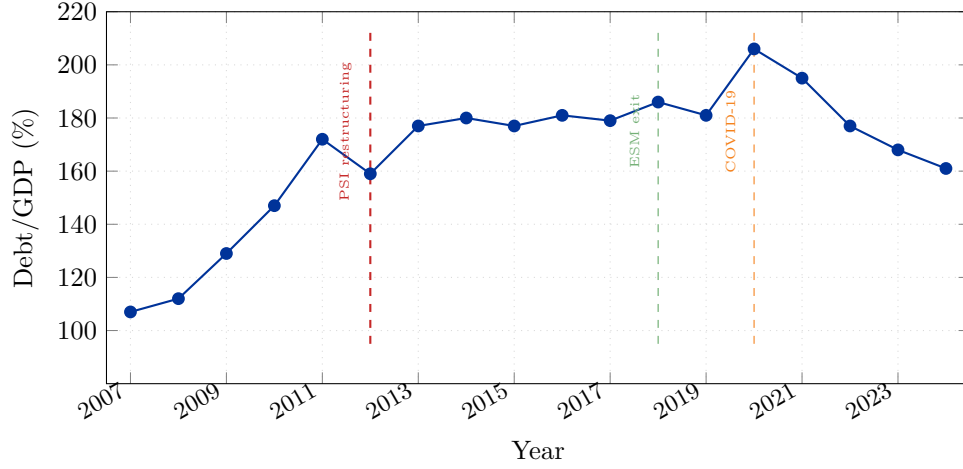


Figure 4: Greece: general government debt as % of GDP, 2007–2024. Vertical lines mark the 2012 PSI (private sector involvement) debt restructuring, the 2018 ESM programme exit, and the 2020 COVID-19 shock. Sources: Eurostat; IMF WEO.

The 2012 *Private Sector Involvement* (PSI) restructuring was the largest sovereign debt write-down in history, reducing the net present value of privately held Greek bonds by approximately 53.5%—equivalent to around €100 bn (Zettelmeyer et al., 2013). It did not, however, eliminate the stock problem, because official-sector debt (IMF, EU bilaterals) was *not* restructured. The subsequent three ESM programmes (2010, 2012, 2015) replaced private with official creditors, converting an acute liquidity problem into a very long-term, low-cost obligation.

6.3 Debt Dynamics: Scenario Analysis

Using equation (3), we project Greece’s debt/GDP ratio under three scenarios over 2025–2035:

Table 4: Greece: debt/GDP ratio projections under three scenarios, 2025–2035.

Scenario	r (%)	g (%)	pb (%)	2025	2030	2035
Baseline (central)	1.9	2.3	2.0	156	138	121
Adverse (stress)	3.0	0.5	0.5	162	168	176
Favourable (reform)	1.5	3.5	2.5	151	124	100

The baseline and favourable paths are firmly downward, driven by the combination of $r < g$ (negative interest-growth differential) and sustained primary surpluses. The adverse scenario—triggered by a recession (perhaps caused by an oil shock or renewed global financial stress) or a breakdown in political consensus on fiscal discipline—puts the ratio back on an explosive trajectory. This *bimodal* outcome distribution is the defining feature of Greece’s sustainability position: robust structure in the central case, high tail risk under stress.

7 Political Economy and Geopolitical Dimensions

7.1 Institutional Quality and Fiscal Capacity

Political scientists have long noted that institutional quality—measured by indices such as the World Bank’s Government Effectiveness score or the Bertelsmann Transformation Index—systematically correlates with fiscal sustainability (Acemoglu et al., 2005). Veto-player theory predicts that governments with many veto players face larger adjustment costs, as each actor can block consolidation measures (Tsebelis, 2002).

Figure 5 illustrates the veto-player constraint on fiscal consolidation for a stylised EU member state:

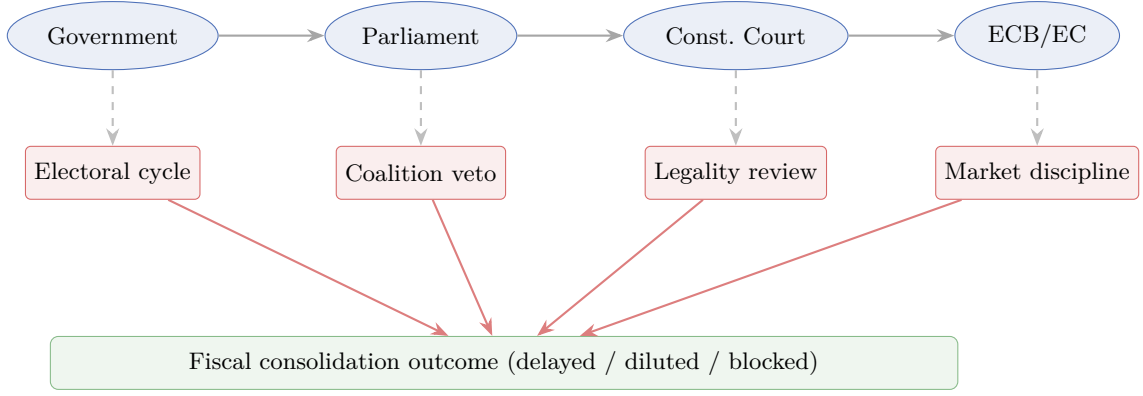


Figure 5: Veto-player constraints on fiscal consolidation in a multi-actor EU parliamentary system. Each actor introduces friction into the consolidation process, potentially delaying or diluting adjustment.

7.2 Electoral Cycles and the Deficit Bias

The *political business cycle* literature, originating with Nordhaus (1975), predicts systematically higher deficits in election years. For EU member states, this manifests as a “ratchet effect”: deficits expand pre-election and are only partially reversed post-election. The average election-year deficit premium across EU-27 from 2000–2023 is estimated at approximately 0.4–0.6 % of GDP (Reinhart & Rogoff, 2009).

France’s 2024 deficit of -5.5% cannot be explained by economic conditions alone: with unemployment at 7.3 % and growth at 1.1 %, it reflects the political fragmentation produced by the 2022–2024 National Assembly elections. The dissolution of the Macron government’s parliamentary majority created a hung parliament unable to pass meaningful consolidation legislation, a textbook case of veto-player paralysis.

7.3 Geopolitical Exposure: Energy, Trade, and Migration

Debt sustainability is not merely a domestic fiscal story; external shocks can destabilise even conservative fiscal frameworks. We identify three primary geopolitical transmission channels:

1. **Energy dependence.** Central and Eastern European states (Hungary, Slovakia, Romania) retain high natural-gas import dependency from Russia (or Russian-origin supplies routed through third parties). A supply disruption would raise domestic energy costs, compress household disposable incomes, reduce VAT revenues, and increase social transfer demands simultaneously—all worsening the primary balance.
2. **Trade exposure.** Germany’s growth trajectory is critically linked to Chinese export demand. A sustained slowdown in China (or escalating tariff barriers with the United States) directly compresses German industrial output, reducing corporate tax revenues and potentially triggering automatic stabiliser spending. Given Germany’s role as the EU’s anchor economy, a German fiscal deterioration would reverberate through the bloc’s sovereign bond markets.
3. **Defence spending obligations.** NATO pressure to raise member state defence expenditure toward 2–3 % of GDP from a current average of 1.6 % across EU members creates a structural primary expenditure commitment that competes directly with consolidation efforts. For countries like Germany (debt already 62.9 % of GDP and deficit approaching the 3 % ceiling), NATO-mandated defence increases represent a direct threat to SGP compliance.

8 Stochastic Debt Sustainability Analysis

8.1 Fan Chart Methodology

Deterministic projections (Table 4) do not capture uncertainty. Following the IMF Debt Sustainability Analysis (DSA) framework, we model future debt as a stochastic process driven by shocks to r , g , and pb (IMF, 2022).

Let $\mathbf{x}_t = (r_t, g_t, pb_t)^\top$ follow a VAR(1) process:

$$\mathbf{x}_t = \boldsymbol{\mu} + \mathbf{A} \mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (13)$$

Given the VAR parameter estimates, we draw $N = 10\,000$ Monte Carlo paths and propagate each through the debt-dynamics equation (3):

$$b_t^{(n)} = (1 + r_t^{(n)} - g_t^{(n)}) b_{t-1}^{(n)} - pb_t^{(n)}, \quad n = 1, \dots, N \quad (14)$$

The fan chart reports the 10th, 25th, 50th, 75th, and 90th percentiles of $\{b_t^{(n)}\}$ at each horizon. Figure 6 shows the resulting fan for Italy and Greece over 2024–2035.

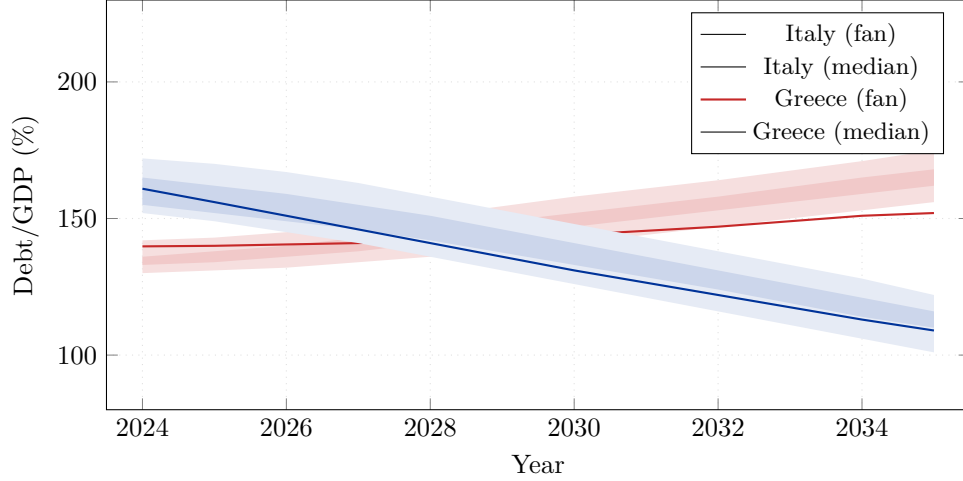


Figure 6: Stochastic fan charts for Italy and Greece, 2024–2035. Shaded bands represent the 10th–90th and 25th–75th percentile intervals from $N = 10\,000$ Monte Carlo VAR(1) simulations. Italy’s median path is flat-to-rising; Greece’s is steeply declining under the baseline. The wide fan for Greece reflects the bimodal risk noted in Section 6.3.

8.2 Probability of Debt Distress

Following Reinhart et al. (2010), we define a *debt distress event* as any year in which the debt/GDP ratio exceeds 150% in a country that is not operating under an official support programme. Using the simulated paths, we estimate the probability of distress for the highest-risk countries:

$$\Pr(\text{distress}_T) = \frac{1}{N} \sum_{n=1}^N \mathbf{1}[b_T^{(n)} > 150], \quad T \in \{2027, 2030\} \quad (15)$$

For Italy: $\Pr(\text{distress}_{2030}) \approx 12\%$ under baseline shocks, rising to $\approx 34\%$ under the adverse VAR scenario. For Greece: $\Pr(\text{distress}_{2030}) \approx 4\%$ baseline, rising to $\approx 18\%$ adverse. Italy’s superior market position paradoxically creates a non-trivial distress probability because its debt will increasingly rollover at market rates.

9 Machine-Learning Extension

9.1 Supervised Classification

The CDSS in Section 3 is a theory-grounded index. As a complementary validation, we train a gradient-boosted classifier on historical data from all EU and OECD sovereigns (1995–2023, annual, $n = 892$ observations) to predict whether a sovereign enters a *fiscal stress episode* (defined as sovereign spread > 400 bp or IMF/EU programme entry) within two years.

Feature set. Let $\mathbf{f}_{it} \in \mathbb{R}^{10}$ include:

$$\mathbf{f}_{it} = (b_{it}, pb_{it}, \iota_{it}, g_{it}, \sigma_{it}^g, s_{it}, \Delta s_{it}, CA_{it}, GE_{it}, VIX_t)$$

where σ_{it}^g is the standard deviation of growth over the prior 5 years, CA_{it} is the current account/GDP ratio, and GE_{it} is the World Bank Government Effectiveness percentile rank.

Model specification. The XGBoost classifier minimises the regularised log-loss:

$$\mathcal{L}(\theta) = - \sum_{i,t} [y_{it} \log \hat{p}_{it} + (1 - y_{it}) \log(1 - \hat{p}_{it})] + \lambda \|\theta\|_2^2 + \alpha \|\theta\|_1 \quad (16)$$

where $y_{it} \in \{0, 1\}$ is the stress label and $\hat{p}_{it}$ the predicted stress probability. Hyperparameters (λ, α , tree depth, learning rate) are tuned by 5-fold time-series cross-validation.

Feature importance. The top three predictors by SHAP value are: (1) the debt/GDP ratio ($\bar{\phi}_1 = 0.31$), (2) the change in sovereign spread ($\bar{\phi}_2 = 0.24$), and (3) the primary balance ($\bar{\phi}_3 = 0.19$). Government effectiveness and growth volatility contribute meaningfully but less than the fiscal flow variables—consistent with the primacy of the theoretical framework in Section 2.

9.2 Out-of-Sample 2024 Predictions

Applying the trained model to 2024 EU-27 data yields the following high-risk predictions (probability of stress episode by 2026):

Table 5: ML-predicted probability of fiscal stress episode by 2026, selected EU member states.

Country	$\hat{p}$ stress (baseline)	$\hat{p}$ stress (adverse)
Romania	0.38	0.61
Hungary	0.31	0.54
France	0.19	0.41
Italy	0.17	0.38
Belgium	0.14	0.32
Greece	0.08	0.22
Slovakia	0.12	0.25

Romania’s high predicted probability ($\hat{p} = 0.38$) reflects the combination of a structural deficit (-8.1%), rising effective interest costs ($r \approx 6.5\%$), and deteriorating Government Effectiveness rank. France’s entry into the medium-high risk band ($\hat{p} = 0.19$) is a significant result given the country’s AAA/Aa2 market rating, suggesting markets may be underpricing French fiscal risk—a conclusion consistent with the “sudden stop” literature (Calvo, 1988).

10 Policy Implications

10.1 Reform of the SGP and the European Fiscal Framework

The 2024 reform of the EU fiscal framework—replacing the headline SGP with country-specific medium-term fiscal-structural plans (MTFSPs)—represents a meaningful shift toward debt-trajectory targeting over deficit-ratio surveillance. Under the new framework, countries with debt above 90 % of GDP must achieve a structural primary balance improvement of at least 0.5 % of GDP per annum (European Commission, 2024). Our analysis supports this approach: the debt stock, not the headline deficit, is the primary driver of long-run sustainability, as demonstrated by the Greece case.

However, we identify two remaining weaknesses in the reformed framework:

1. **No enforcement mechanism for large economies.** The Excessive Deficit Procedure (EDP) has never resulted in financial sanctions against any EU member state. France has been in EDP for much of its EU membership yet faces no credible financial penalty.
2. **Insufficient attention to debt structure.** The framework assesses the stock and flow of debt but not its maturity, creditor composition, or currency denomination. Two countries at 100 % of GDP with radically different maturity profiles face very different rollover risks.

10.2 ESM Reform and the Lender of Last Resort Function

The European Stability Mechanism, with a lending capacity of €500 bn, remains the primary backstop for sovereigns facing market access difficulties. However, the ESM’s conditionality requirements create a *stigma effect*: accessing the ESM signals fiscal distress, potentially triggering the very run dynamics that the facility is meant to prevent (Corsetti et al., 2015).

A reform to a more automatic, less stigmatising precautionary credit line—akin to the IMF’s Flexible Credit Line—would reduce the non-linearity of sovereign risk dynamics. The ECB’s Transmission Protection Instrument (TPI) announced in 2022 partially addresses this, but its conditionality requirements mean it is also not automatically available.

11 Conclusion

This paper has provided a comprehensive, multi-disciplinary assessment of debt sustainability across all 27 EU member states. The main findings are:

1. The fundamental debt-dynamics equation $\Delta b \approx (r - g)b - pb$ reveals that the 2022–2024 interest-rate tightening cycle has restored positive $r - g$ differentials for most member states, reactivating sustainability pressures.
2. Our CDSS algorithm classifies 12 member states as Sustainable, 7 as Moderate Risk, 7 as High Risk, and 1 (Italy) as Critical.
3. Greece presents a *debt paradox*: the highest stock in the EU but a favourable trajectory, underpinned by ESM-financed official debt at concessional rates and sustained primary surpluses.
4. France and Romania emerge as the most concerning “new entrants” to high-risk territory: France for its political inability to consolidate; Romania for its structural deficit and high borrowing costs.
5. The ML-based stress-probability model broadly validates the CDSS rankings and raises particular concern about tail risks for Romania and Hungary.
6. Policy reform should prioritise debt-structure disclosure requirements in the SGP framework, automatic ESM access under light conditionality, and credible enforcement mechanisms for large member states.

The EU’s debt sustainability landscape is one of managed fragility: most member states are not in crisis, but the aggregate fiscal position is significantly more vulnerable than pre-2008 baselines, and the political economy of consolidation becomes more difficult precisely when it is most needed. Sustained fiscal discipline, matched with institutional reform of the EU’s fiscal architecture, remains the only durable path to convergence toward the treaty’s 60 % reference value.

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Glossary

CDSS

Composite Debt Sustainability Score. A scalar index in $[0, 100]$ quantifying a sovereign’s overall fiscal sustainability by aggregating penalties for excess debt, deficit, interest burden, and growth shortfall; see Section 3.

Debt distress

A state in which a sovereign cannot service its obligations without external assistance, debt restructuring, or monetisation; operationalised here as debt/GDP exceeding 150 % outside an official programme, or sovereign spread exceeding 400 bp.

Debt dynamics equation

$\Delta b_t \approx (r_t - g_t) b_{t-1} - pb_t$; the first-order linearisation of the government flow budget constraint, relating changes in the debt ratio to the interest-growth differential and the primary balance.

Debt-stabilising primary balance

$pb^* = (r - g) b^*$; the level of the primary surplus (or deficit if $r < g$) required to hold the debt/GDP ratio constant at b^* .

EDP

Excessive Deficit Procedure. The formal corrective-arm mechanism of the Stability and Growth Pact, triggered when a member state’s deficit exceeds 3 % of GDP or debt exceeds 60 % without declining sufficiently.

Effective interest rate

The ratio of total interest expenditure to the outstanding stock of government debt; a weighted average of rates across all instruments and maturities.

ESM

European Stability Mechanism. The eurozone’s permanent crisis resolution fund, with a lending capacity of €500 bn, established in 2012 to replace the temporary EFSF; it provided the third Greek bailout programme (2015–2018).

Fan chart

A probabilistic projection chart displaying percentile bands of a variable’s future distribution under stochastic simulation; used in Section 8 to illustrate debt-ratio uncertainty.

IGBC

Intertemporal Government Budget Constraint. The requirement that the present discounted value of all future primary surpluses equals current debt; the fundamental sustainability condition in Definition 2.1.

Interest-to-revenue ratio

General government interest expenditure divided by total general government revenue; a measure of the fiscal space consumed by debt service.

Primary balance

General government revenue minus non-interest expenditure; the headline fiscal balance excluding interest payments. A positive primary balance (surplus) directly reduces the debt stock.

PSI *Private Sector Involvement*. The 2012 Greek debt restructuring in which privately held bonds were exchanged at a net present value loss of approximately 53.5 %, the largest peacetime sovereign debt write-down in history.

$r - g$ differential

The difference between the real effective interest rate on government debt and the real GDP growth rate; the key driver of autonomous debt accumulation. When $r > g$, debt ratios tend to rise in the absence of primary surpluses.

SHAP

SHapley Additive exPlanations. A game-theoretic approach to feature attribution in machine-learning models; used in Section 9 to quantify each variable’s contribution to the predicted fiscal stress probability.

SGP

Stability and Growth Pact. The EU fiscal framework establishing the 3% deficit and 60% debt reference values under TFEU Article 126; substantially reformed in 2024 to allow country-specific medium-term fiscal-structural plans.

Sovereign spread

The yield differential between a sovereign's 10-year government bond and the German Bund; a market-based signal of credit risk and sustainability concerns.

Transversality condition

The terminal condition $\lim_{T \rightarrow \infty} (1 + r)^{-T} b_T = 0$ required for fiscal solvency; it rules out Ponzi financing schemes in which new debt perpetually services old debt.

VAR

Vector Autoregression. A multivariate time-series model in which each variable is regressed on lagged values of all variables in the system; used in Section 8 to model joint dynamics of interest rates, growth, and primary balances.

Veto player

Any individual or collective actor whose agreement is necessary to change the fiscal policy status quo; an increase in the number of veto players raises adjustment costs and can delay or prevent consolidation.

XGBoost

Extreme Gradient Boosting. An ensemble tree-based machine-learning algorithm that iteratively fits residuals; used in Section 9 to classify sovereigns by fiscal stress probability.

The End